

Feedback-Loopers

Team members:

Ashutosh Kumar (LEAD , **B I**)

Archana Kumari (member ,**B**)

Team background

Academic Experience:

- **B:** B.Tech
- Work Experience
 - **I:** Experience on industry

Feedback-Loopers

Project Title: single-stage voltage-mode WTA circuit for selective attention based processing in massive parallel sensing applications

- **Our goal** is to implement single stage Winner Take All circuit(**WTA**).
- In the **MOSBius** context, we aim to implement single stage so in future. we can aim for multi-stage WTA circuit .

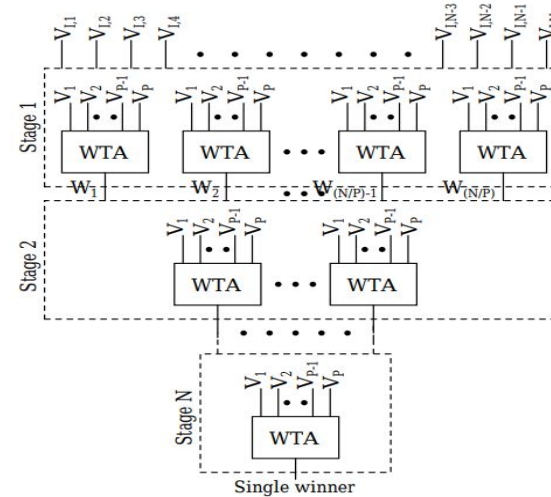


Fig. 1. Multi-stage WTA approach

Image taken from [1] P. K. Pandey, B. Bhuvan: A 1024-Input Multi-Stage Voltage-Mode WTA Circuit for Selective Attention Based Processing in Massive Parallel Sensing Applications. iSES 2024: 49-53

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Basic Architecture of WTA.

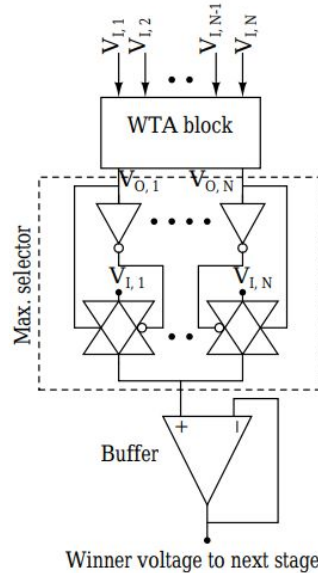


Fig. 3. WTA with max selector and buffer

Image taken from [1] P. K. Pandey, B. Bhuvan: A 1024-Input Multi-Stage Voltage-Mode WTA Circuit for Selective Attention Based Processing in Massive Parallel Sensing Applications. iSES 2024: 49-53

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Design approach of WTA :

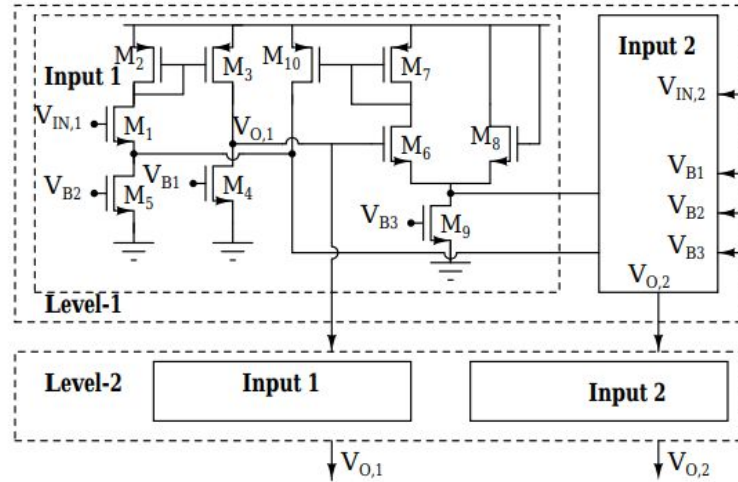


Fig. 2. Two-level two-input WTA

Image taken from [1] P. K. Pandey, B. Bhuvan: A 1024-Input Multi-Stage Voltage-Mode WTA Circuit for Selective Attention Based Processing in Massive Parallel Sensing Applications. iSES 2024: 49-53

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Testing

Approach

:

- **Inputs are applied from outside.**
- Output will be tested with winner value.

Feedback-Loopers

Advantages

The circuit uses multi-stage WTAs to improve the resolution and speed.

Application:

- Used in Sensor arrays, edge AI, complex neural networks and nonlinear filters for image processing applications.

References:

- [1] P. K. Pandey, B. Bhuvan:**A 1024-Input Multi-Stage Voltage-Mode WTA Circuit for Selective Attention Based Processing in Massive Parallel Sensing Applications.** iSES 2024: 49-53
- J. Choi and B. Sheu, "A high-precision vlsi winner-take-all circuit for self-organizing neural networks," IEEE Journal of Solid-State Circuits, vol. 28, no. 5, pp. 576–584, 1993
- J. Lazzaro, S. Lyckebusch, M. A. Mahawold, and C. Mead, "Winnertake-all networks of $O(n)$ complexity," in Advances in Neural Information Processing Systems, D. S. Touretzky Ed., pp. 703–711, Morgan Kaufmann, 1989.

Schematic

Top Level: WTA8_1_tb

```

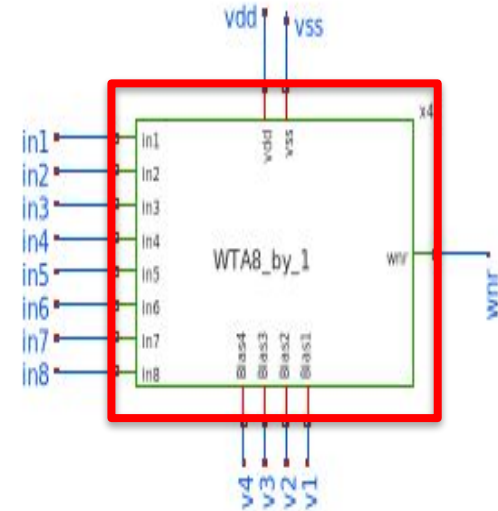
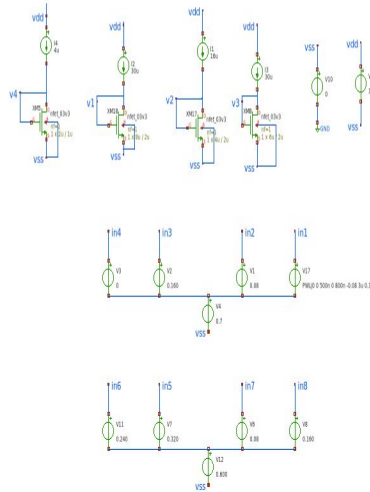
STIMULI
.option savecurrents

.control
save all
OP
*DC VS 0 3.3 0.01
tran 50p 3u
*remzerovec
write test_WTA.raw

.endc

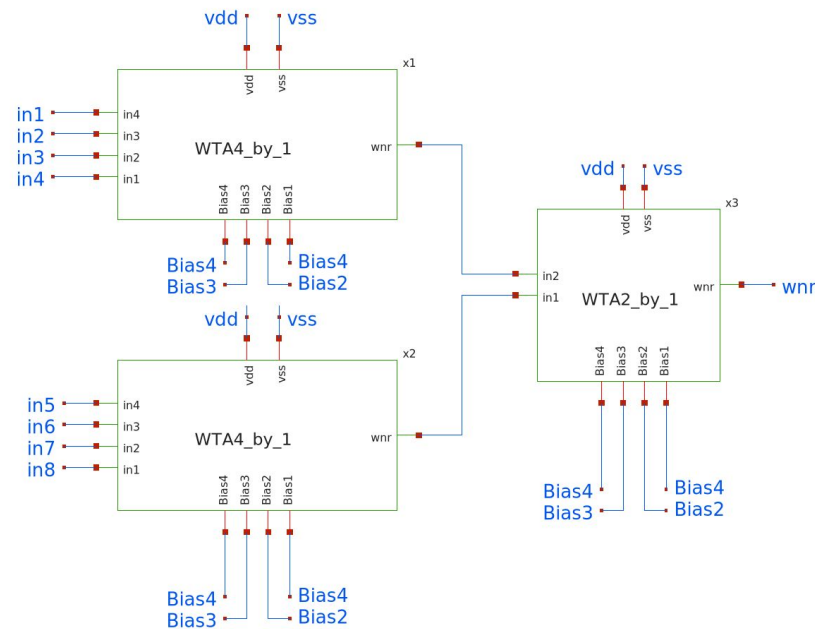
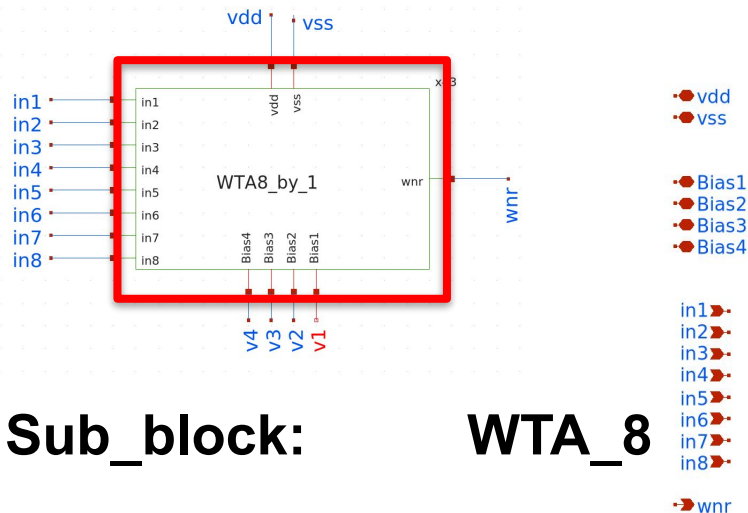
```

```
MODELS
.include $::180MCU_MODELS/design.ngspice
.lib $::180MCU_MODELS/sw141064.ngspice typical
.include $::180MCU_MODELS/design.ngspice
```



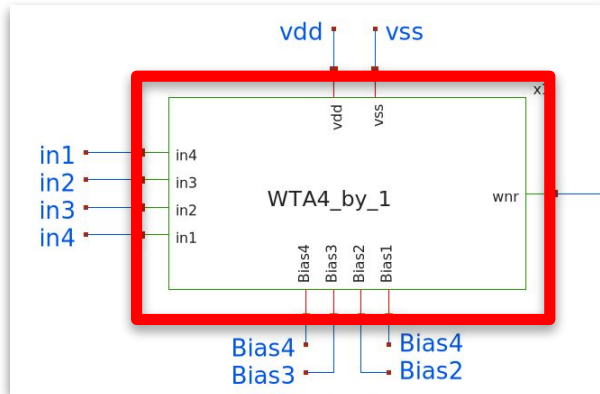
Top_Block

Schematic

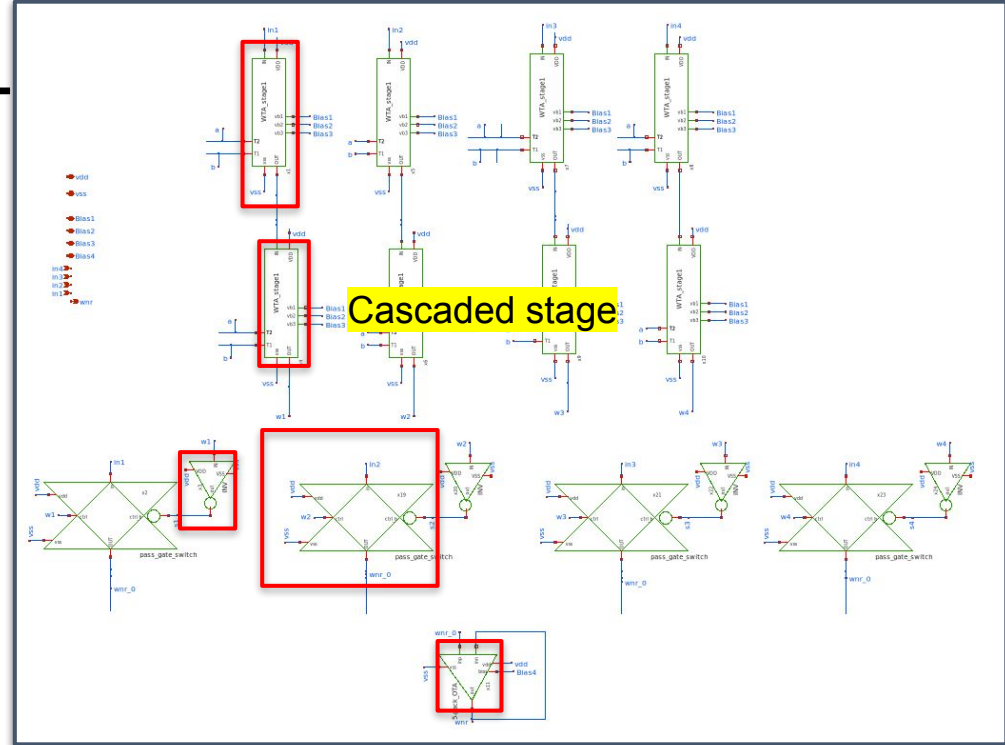


Schematic

Top

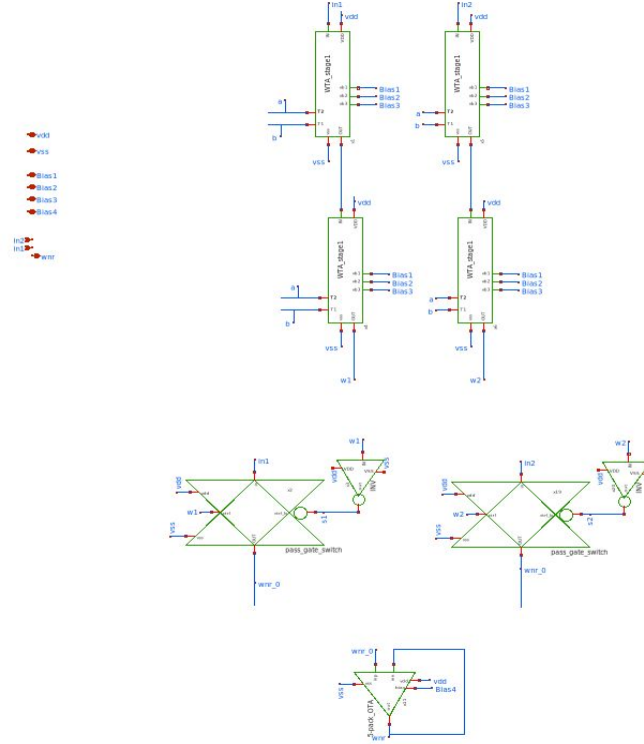
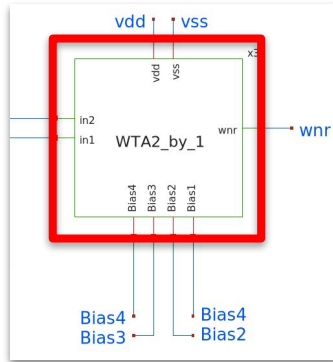


L



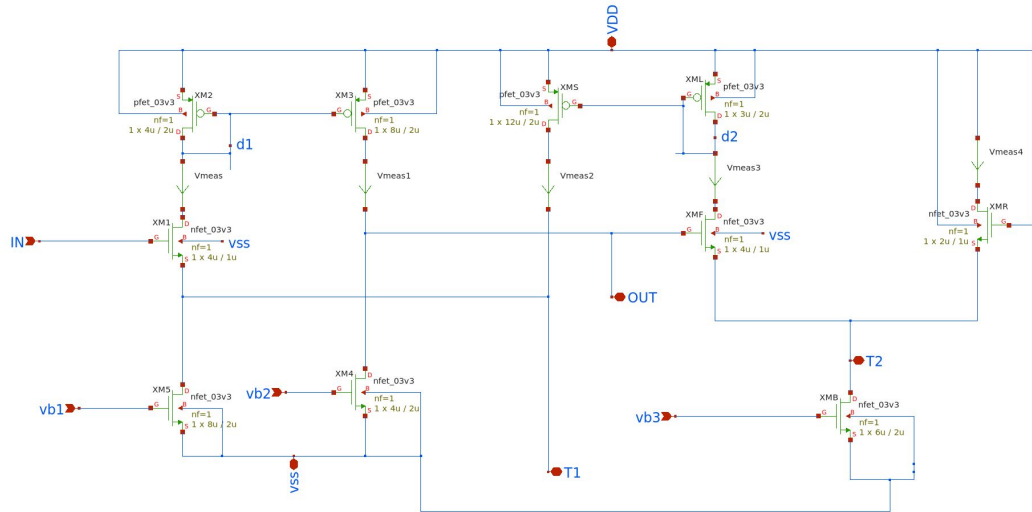
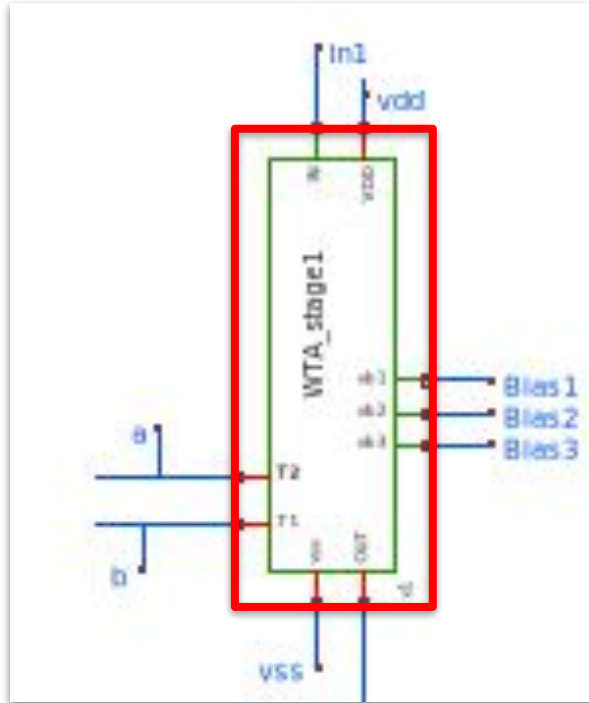
Schematic

Sub_block: WTA_2:1



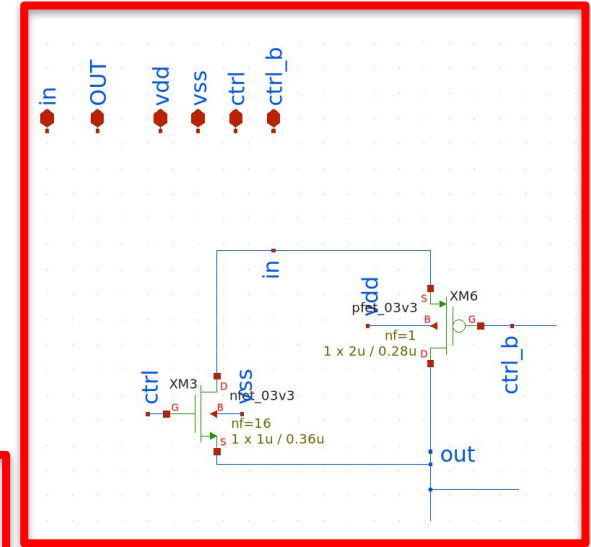
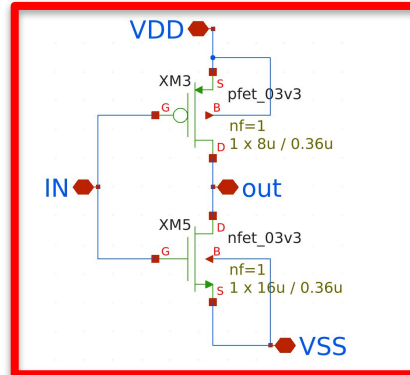
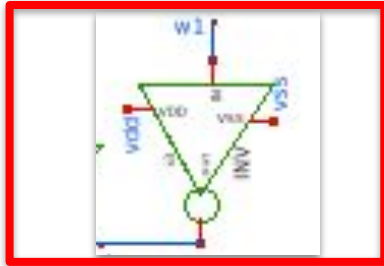
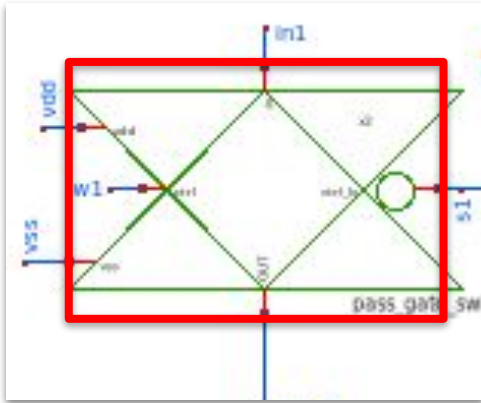
Schematic

Sub_block: WTA single stage



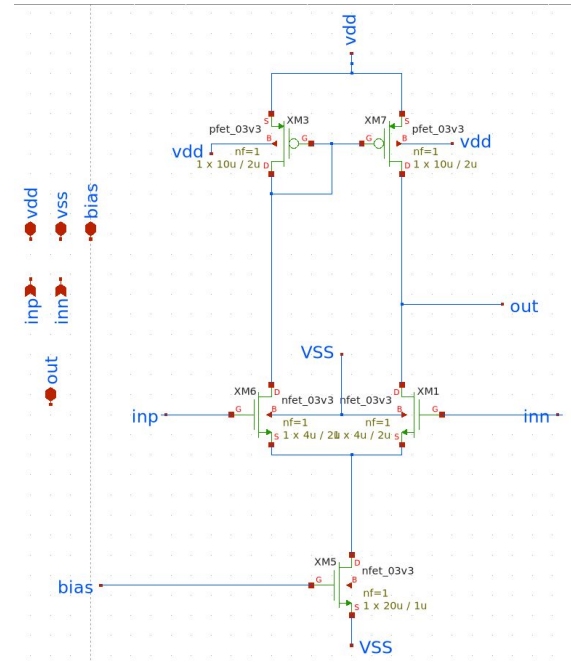
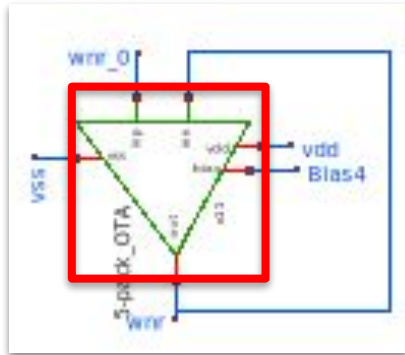
Schematic

Sub_block: Transmission_gate and Inverter



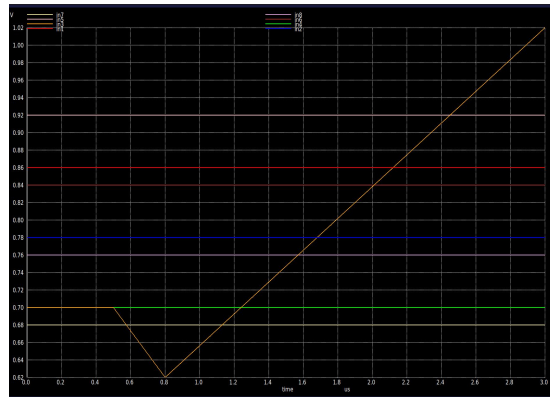
Schematic

Sub_block: 5pack_OTA -> Buffer

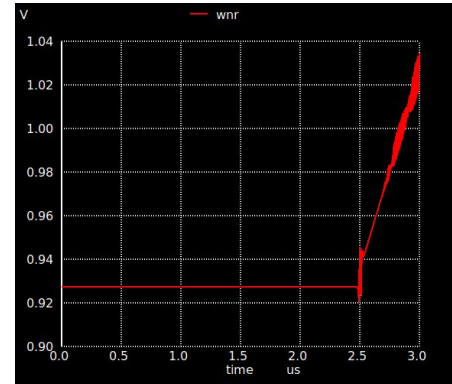


Simulation -Results

- Highest input : 0.92v and in3 ramping up
- OUTPUT : track highest value(winner)



INPUT

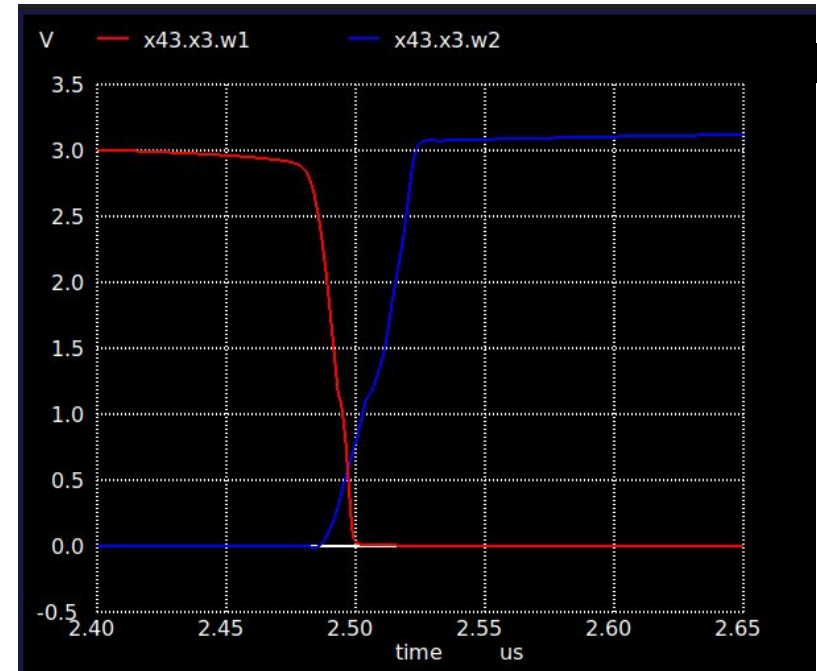


OUTPUT(Winner)

Simulation -Results

OUTPUT : WTA_2:1

- As input
Winner selection changes.



Pin -In/out & Area

- **Pin : 8 input , 4 bias , 1 output**
- **Area : $\sim 170 \times 70 \mu\text{m}$ (estimated by instantiating all pcells)**

Progress-track sheet

Hierarchical Level	Task	%	Schematic	Progress	Pre-PEX SIM	Progress	layout-LVS	layout - DRC	Post-PE X SIM	Progress
1	WTA_8:1	80%			90%		Fail	Fail	0%	
2	WTA_4:1	0%			90%				0%	
3	WTA_2:1	80%			90%				0%	
3A	WTA_single_STAGE	80%			90%				0%	
3B	Trasmission_gate	0%			90%				0%	
3C	Inverter	0%			90%				0%	
3D	OTA_buffer	0%			90%				0%	
3	Calibration and Optimization	0%			80%				0%	
3	Padframe intregation	0%			0%				0%	
3	Full-chip Simulation	0%			0%				0%	
final	Tape-out Checklist	0%			0%				0%	
final	Documentation	0%			50%				0%	

Link : [tracking sheet](#)

Current status

- Top pre-simulation -done , optimisation needed.
- Preliminary -output looks good .
- Layout need to be done.
- **Seeking help** regarding biasing of current mirror.